

I. Background and Problem Statement:

1.1 Project background

Dyslexia is a common neurodevelopmental disorder characterized by difficulties with literacy, reading, and spelling that cannot be explained by intellectual, educational, or sensory problems.^[1] This is due to the brain's congenital disorders in the conversion of "glyphs→phonetics→semantics", rather than personal intelligence or learning attitudes.

The disorder occurs in all age groups worldwide and has a wide impact. In learning and working, people with dyslexia face significant challenges: students are inefficient in reading textbooks and completing homework; Adults have difficulty understanding when dealing with literature and work documents, which affects the quality of learning and work efficiency.^[2]

In the digital age, most textual content is not tailored to the needs of the dyslexic population. Global research on dyslexia assistance is still in the early stages, leading to a notable gap in the development of professional and targeted auxiliary tools for this group.

1.2 Core questions

The core issues faced by the current group of individuals with reading difficulties can be summarized into the following four categories^[1]:

(1) Difficulty in character recognition: Easily confusing characters with similar shapes (such as 'b' and 'd', '±' and '士'), or reversing the order of words and sentences, leading to obstacles in basic reading^[3].

(2) Low reading fluency: Needing to read word by word, unable to achieve "whole word recognition," resulting in slow reading speed, frequent rereading, and easily distracted attention, making it hard to focus on text comprehension.

(3) Obstacles derived from spelling and writing: Difficulty in recognizing characters directly leads to a high rate of spelling errors, while a weak understanding of the phonetic and semantic associations of complex vocabulary affects information absorption.

(4) Accompanied by cognitive challenges: Some patients have weak working memory, easily distracted attention, and may experience psychological stress and a sense of avoidance when facing large blocks of text.

1.3 Limitations of existing solutions

There are already some reading assistance tools available in the market, such as Helperbird^[4], Microsoft Immersive Reader^[5], Kurzweil 3000^[6], etc., but these tools generally have the following shortcomings:

(1) Single-function focus: Most tools only focus on visual optimization (such as adjusting font size, background color), lacking processing of the text content itself (such as splitting long sentences, replacing complex vocabulary), making it difficult to fundamentally reduce the semantic understanding barrier for readers with reading difficulties;

(2) Limited scenario coverage: Existing tools are mostly suitable for general reading scenarios, lacking targeted support for specific scenarios such as academic literature, student assignments, etc., and some tools lack multimodal support (such as synchronized voice following and text highlighting);

(3) Low level of personalization: Different types of reading difficulties (surface-level, deep-level, phonetic) were not considered, making it difficult to provide customized support solutions.

1.4 Project Significance

This project aims to develop a multimodal adaptive reading assistance system for individuals with dyslexia, addressing the core reading pain points of those with dyslexia from a technical perspective. The implementation of this project holds three key significances:

(1) Social significance: Focus on the information access rights of people with dyslexia, a vulnerable group, narrow the digital divide, and promote inclusive social development

(2) Academic significance: Combining technologies such as NLP and large language models with the cognitive psychology characteristics of dyslexia, exploring innovative development models for digital assistive tools, providing practical references for research in related fields;^[7]

(3) Practical significance: Providing efficient, convenient, and personalized reading assistance solutions for individuals with dyslexia, covering multiple scenarios such as student learning and adult work, enhancing their learning and work efficiency, and strengthening their sense of social participation.^[8]

II.Objectives and Outcome

2.1 Project objectives

This project will develop a dyslexia-assisted reading system that integrates text optimization, visual adaptation, and multi-modal assistance to solve the core problems of dyslexia in glyph recognition, reading fluency, semantic understanding, etc., and improve their reading experience and efficiency. The specific objectives are as follows:

(1) Realize the intelligent optimization function of text^[9]: support long sentence splitting, short sentence merging, and complex vocabulary replacement to ensure that the length of a single sentence conforms to the cognitive habits of dyslexics (no more than 15-20 words in Chinese and no more than 12 words in English).

(2) Provide personalized visual adaptation solutions: built-in dyslexia-friendly fonts, support word spacing, line spacing, and font size adjustment, and provide high-contrast color matching schemes to reduce the burden of glyph recognition;

Develop multimodal auxiliary tools: implement functions such as text-to-speech, keyword annotation, and vocabulary phonetic example query to meet the learning habits of people with different dyslexias.

(3) Adapt to multiple scenarios and languages: Support the import of common document formats such as PDF, adapt to mainstream languages such as Chinese and English, and cover scenarios such as academic reading and daily learning.

(4) Optimize the system interaction experience: adopt a minimalist interface design to reduce interference elements, and the operation steps are no more than 3 steps, lowering the threshold for user use.

2.2 Expected Outcome

2.2.1 Product results

Runnable auxiliary reading system prototype (including front-end interface and back-end service), supporting web access;

System function document: Detail the functional modules, operation process, technical parameters, etc. of the system;

User Manual: A simple operation guide for people with print disabilities, including feature introductions, configuration suggestions, and more.

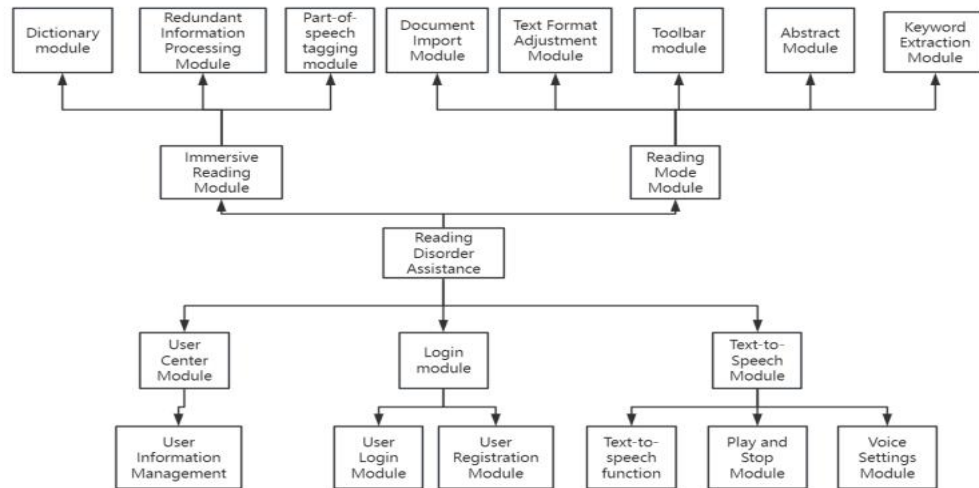
2.2.2 Technological achievements

- (1) **Text optimization algorithm model:** long sentence splitting and vocabulary replacement algorithm developed based on spaCy, HanLP and other tools;
- (2) **Multimodal data interaction interface:** realize efficient interaction of text, voice, visual configuration and other data;
- (3) **System source code and open source documentation:** Ensure the reusability and scalability of the project.

2.3 Project Contributions

- (1) **Technological innovation:** Integrate NLP and large language model technology to achieve intelligent optimization of text content, breaking through the limitations of existing tools that only focus on visual adjustments.
- (2) **Literacy Adaptability:** Conduct special optimization for the reading and writing needs of dyslexic individuals, and fit their core literacy usage scenarios.
- (3) **Practical value:** Provide a "one-stop" reading aid solution that takes into account personalization and ease of use, and can be directly applied to the learning and work of people with dyslexia;

Overview of the top-level requirements of the system's figure as follow:



Picture 1 - top-level requirements of the system

III. Project Methodology

3.1 Literature Review

Dyslexia is a typical neurodevelopmental disorder, the core cause is abnormal brain information processing, which has nothing to do with IQ^[1], visual deficits or acquired